EO-based Downscaling for Air Quality Applications

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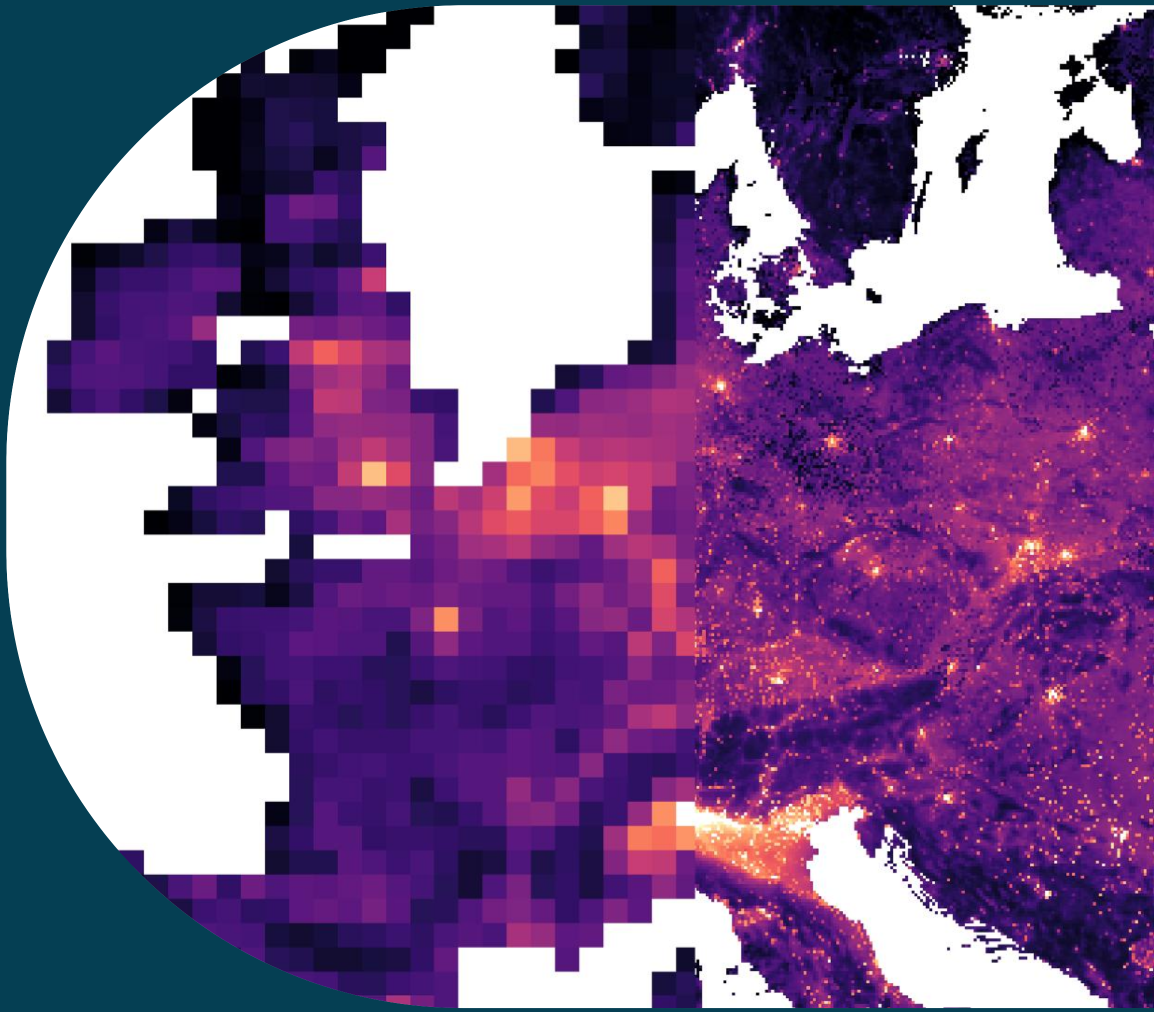

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Introduction

Objective: Make satellite observations of NO2 and AOD more directly applicable for societal applications (e.g. exposure assessment).

Approach: Perform (simultaneous) column to surface conversion and downscaling to finer resolution.

Three examples:

Method 1: TROPOMI NO_2 TVCD to surface conversion and downscaling using high-resolution CTM and geostatistics	**Local scale**
Method 2: TROPOMI NO_2 TVCD to surface conversion and downscaling using Machine Learning Method 3: CAMS $PM_{2.5}$ downscaling using VIIRS AOD/TROPOMI AI and other data using ML	**Continental and local scale**

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Method 1

Local CTM + Geostatistics

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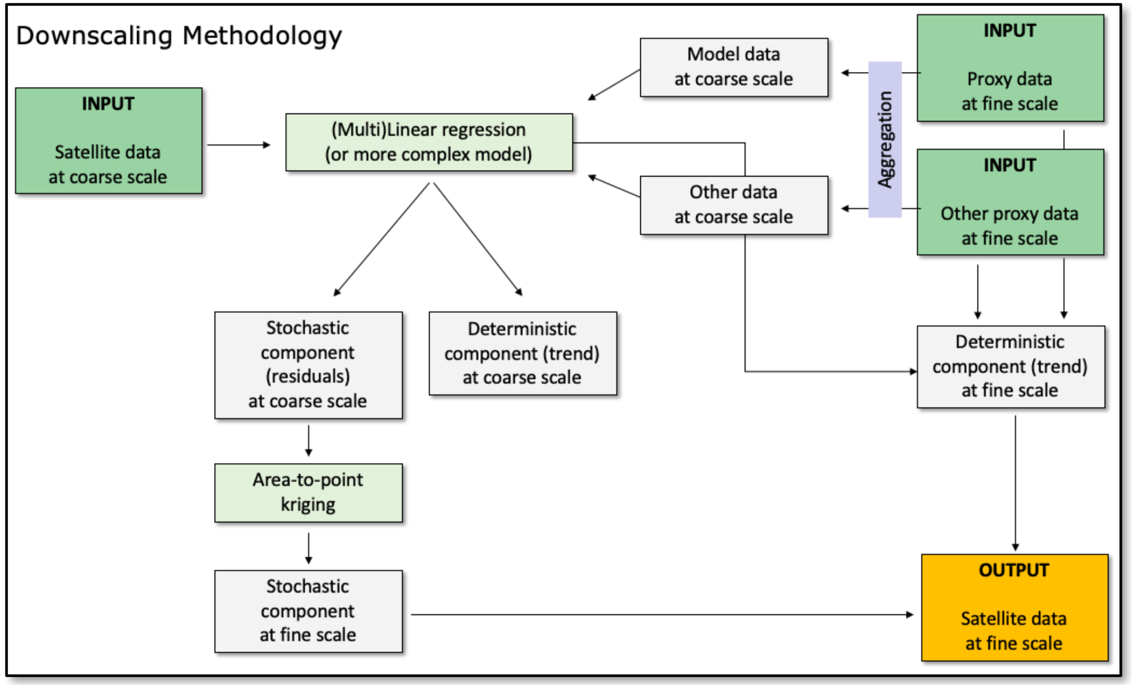
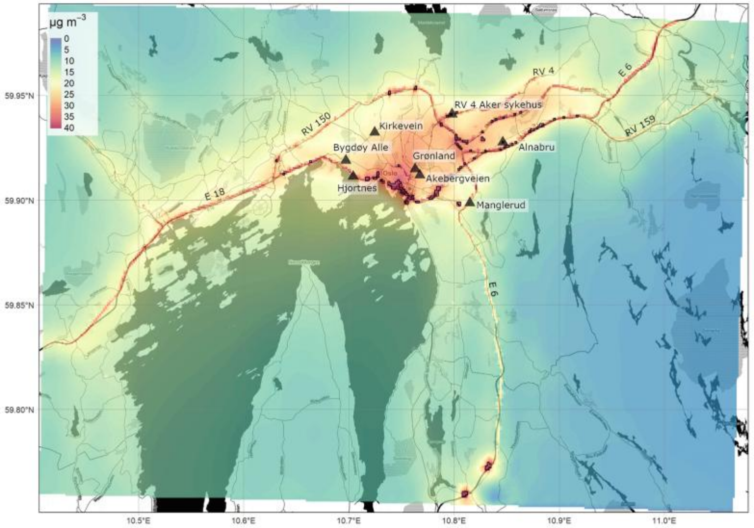
Method 1

- Geostatistical downscaling allows for increasing the spatial resolution of the S5P NO_2 data by exploiting the spatial patterns from a high-resolution proxy dataset
- We use the **local-scale CTM EPISODE** (Hamer et al. 2020) for auxiliary information
- We have developed an **observation operator** to compute EPISODE-based TVCDs **using the TROPOMI Averaging Kernel**
- We derive **surface NO_2 concentration** using a model-based column-to-surface ratio
- Then the **surface NO_2 dataset is downscaled** using residual area-to-point kriging (Kyriakidis, 2004)
- The downscaled result is **mass-conservative**





Annually averaged NO_2 concentrations (µg m^{-3}) from the EPISODE model over Oslo (100 m × 100 m horizontal resolution). The black triangles indicate the locations of air quality observation stations.

Hamer, P. D., Walker, S.-E., Sousa-Santos, G., Vogt, M., Vo-Thanh, D., Lopez-Aparicio, S., Schneider, P., Ramacher, M. O. P., & Karl, M. (2020). The urban dispersion model EPISODE v10.0 – Part 1: An Eulerian and sub-grid-scale air quality model and its application in Nordic winter conditions. *Geoscientific Model Development*, *13*(9), 4323–4353.

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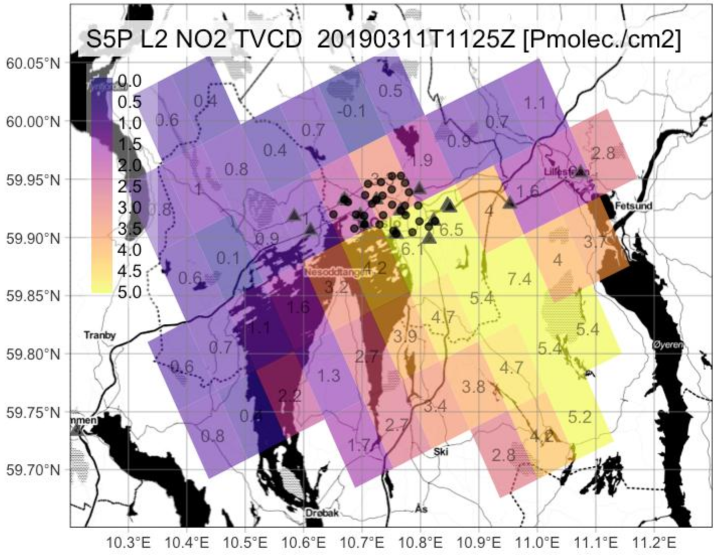
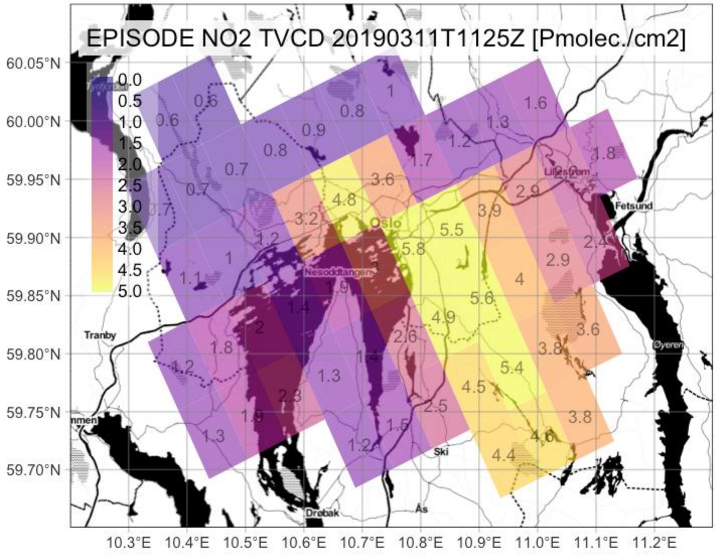
From TVCD to surface NO_2

TROPOMI TVCD

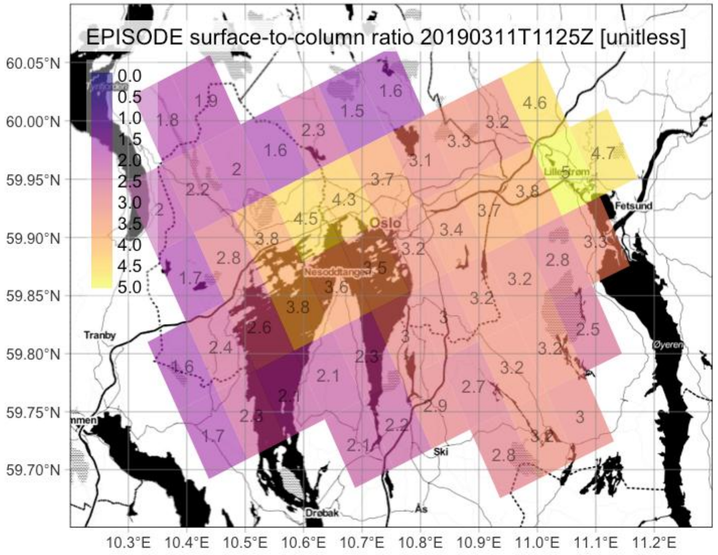


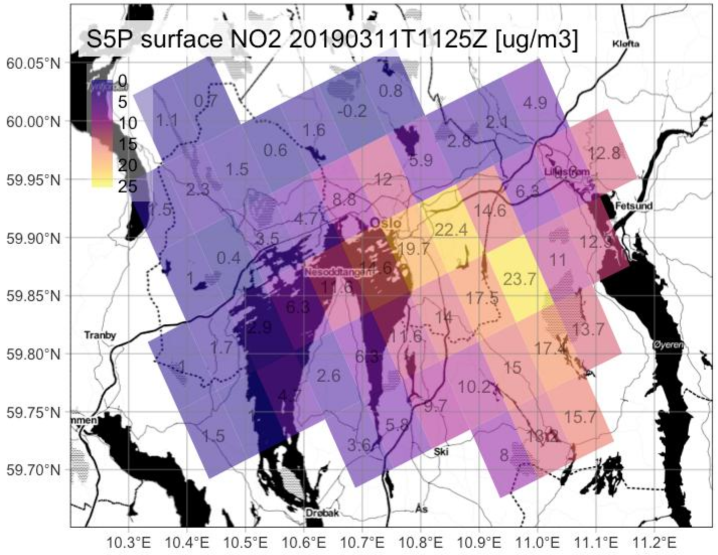


EPISODE TVCD

Conversion of TROPOMI Level-2 TVCD (in Pmolec./cm^2) to surface NO2 (ug/m3)at original satellite footprint using auxiliary info from a physical model (EPISODE)

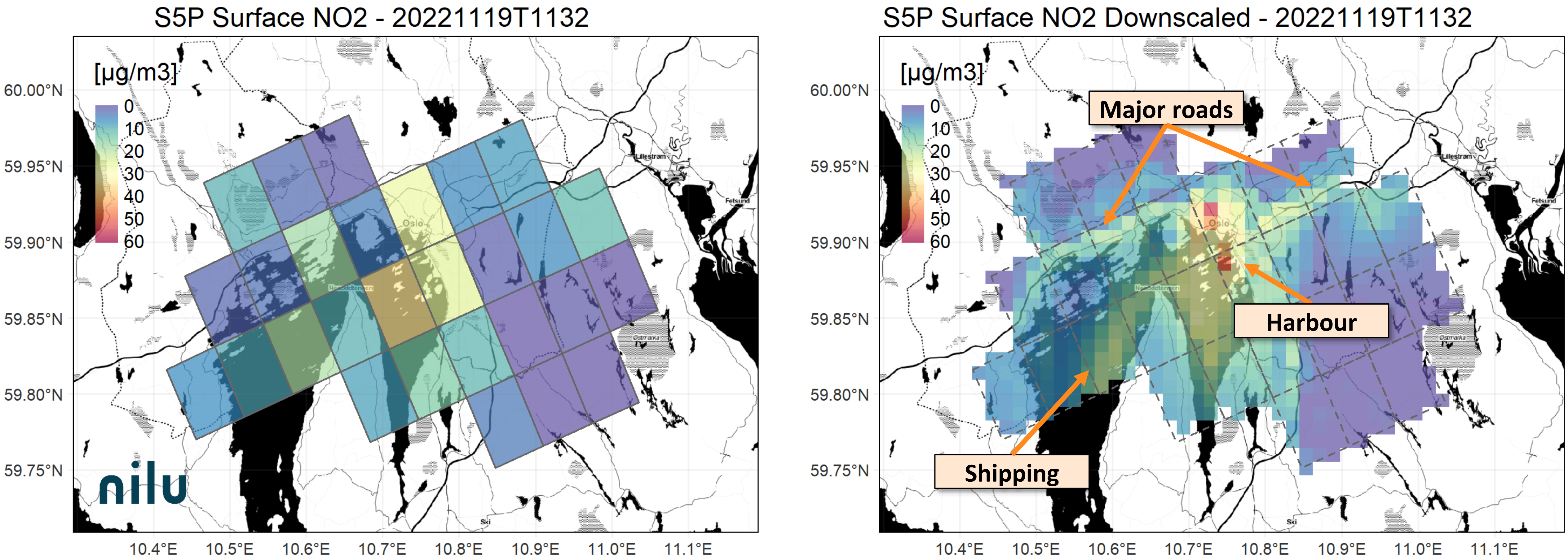
EPISODE surface-to-TVCD ratio





TROPOMI surface NO_2

Downscaling TROPOMI surface NO_2: Oslo test site



TROPOMI-derived surface NO_2 at original Level-2 satellite footprint.

Downscaled TROPOMI-derived surface NO_2 at 1000 m spatial resolution.

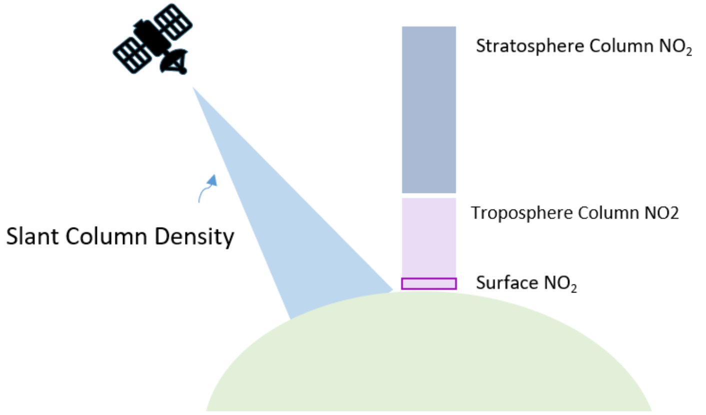
Geostatistical downscaling of TROPOMI-derived surface NO_2 to 1 km resolution, here for 2022-11-19 at 11:32 UTC.

Method 2

ML-based downscaling of TROPOMI NO2

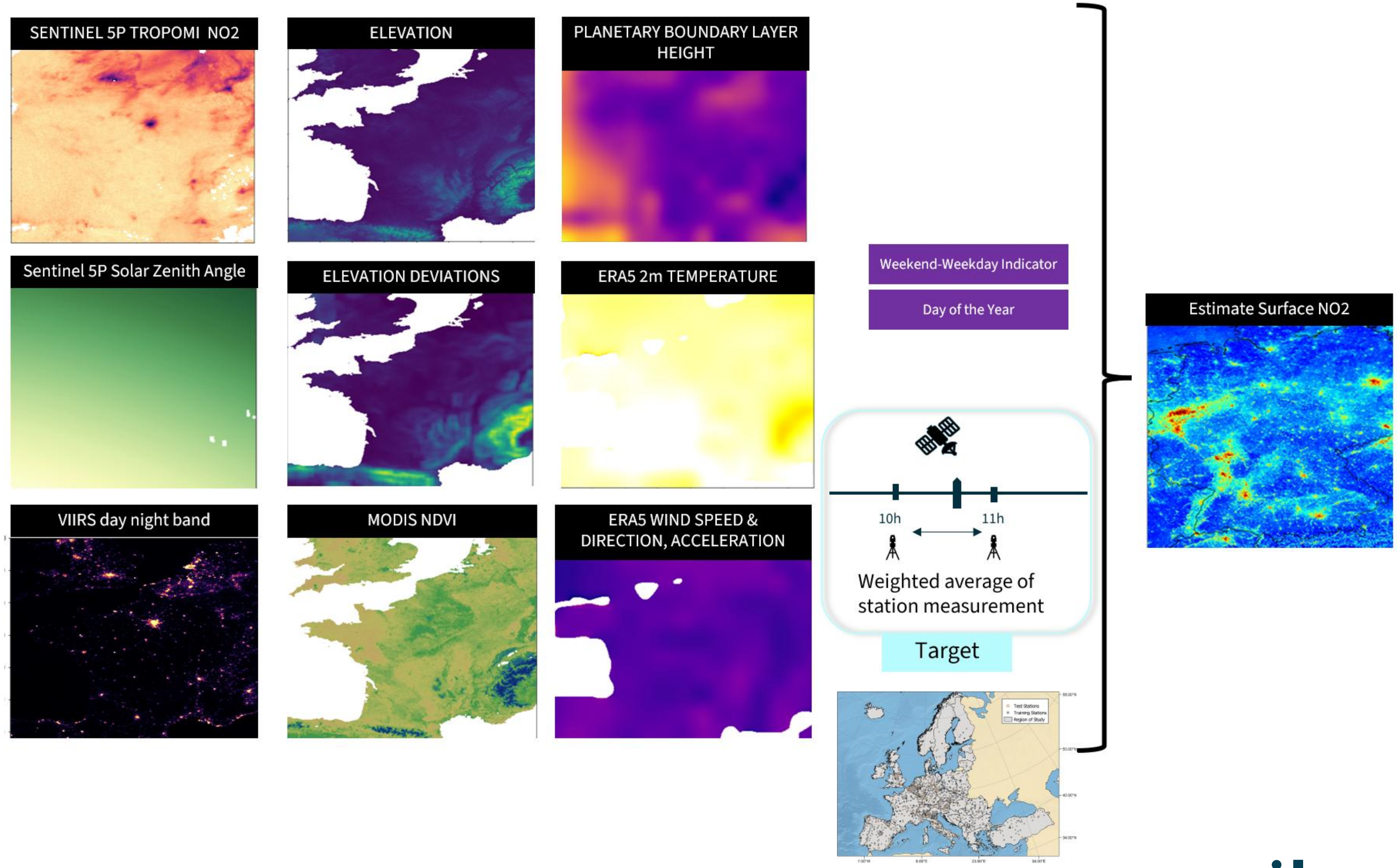
S-MESH: **S**atellite and **M**L-based **E**stimation of **S**urface air quality at **H**igh resolution

Challenge



Approach

Train a European-scale XGBoost model to convert satellite **column to surface NO_2** and **downscale to 1000 m** resolution using reference station data for training



Shetty, S., Schneider, P., Stebel, K., Hamer, P. D., Kylling, A., & Berntsen, T. K. (2024). Estimating surface NO_2 concentrations over Europe using Sentinel-5P TROPOMI observations and Machine Learning. Remote Sensing of Environment, 312, 114321.

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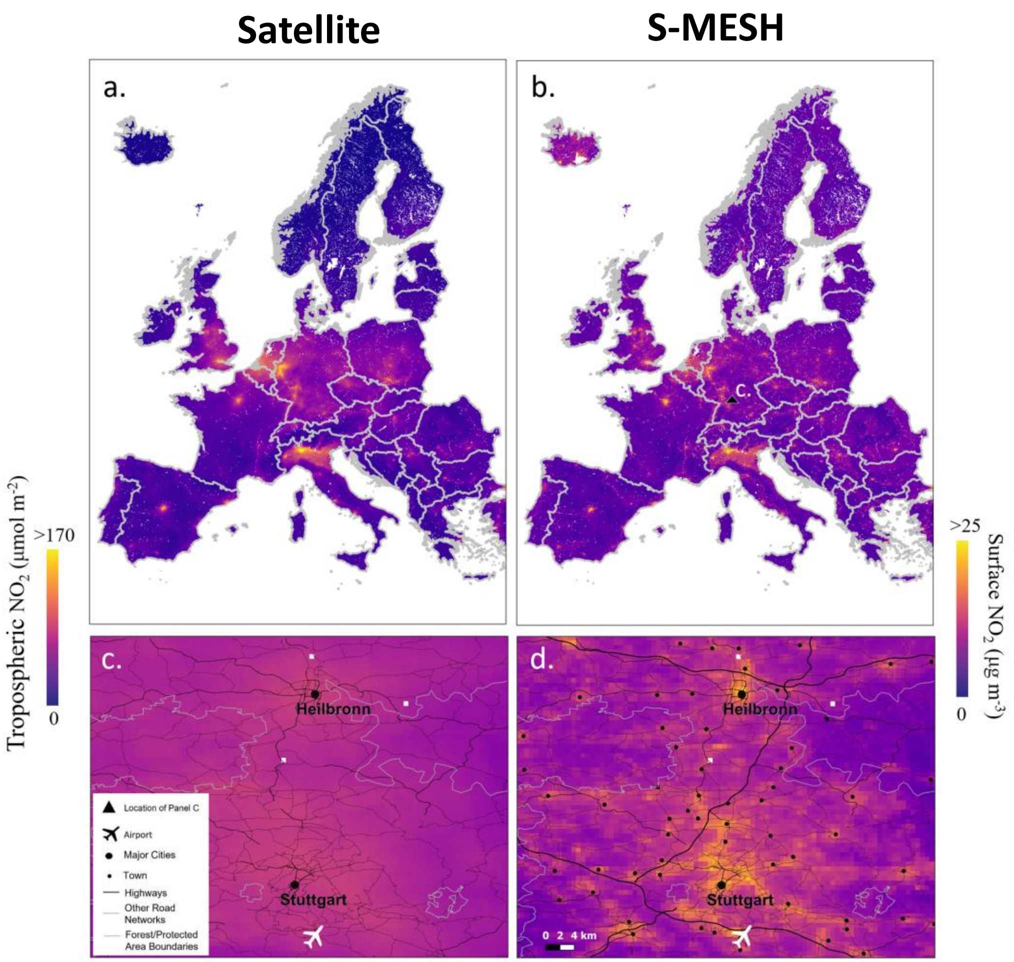
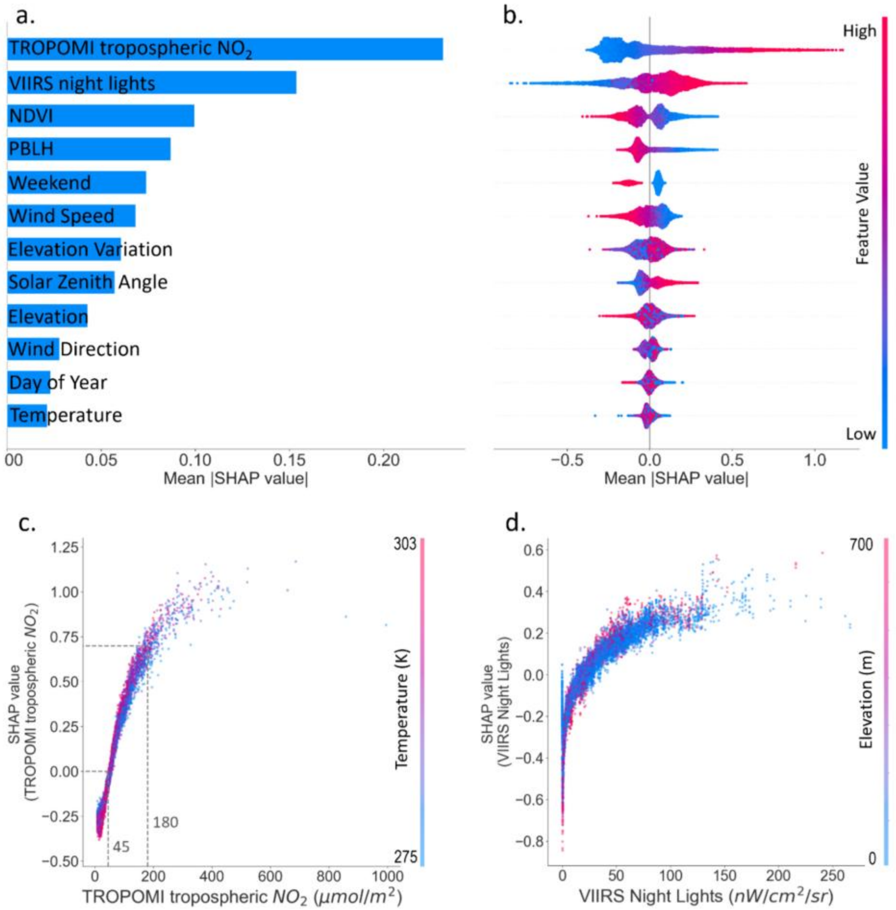


Simultaneous **column-to-surface NO_2 conversion** and **downscaling**



Use of **explainable AI techniques** for better understanding the model

Shetty, S., Schneider, P., Stebel, K., Hamer, P. D., Kylling, A., & Berntsen, T. K. (2024). Estimating surface NO_2 concentrations over Europe using Sentinel-5P TROPOMI observations and Machine Learning. Remote Sensing of Environment, 312, 114321.

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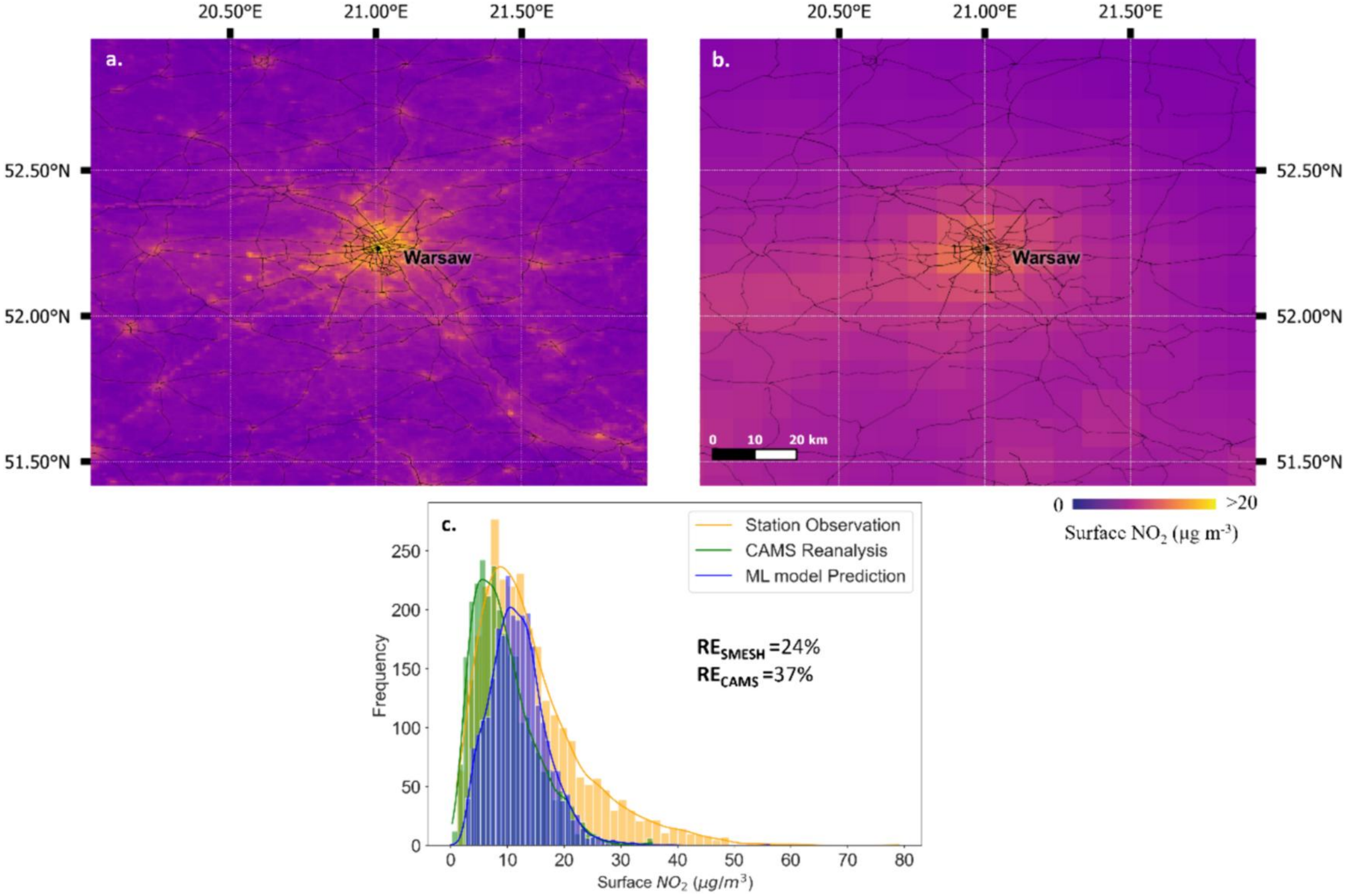
S-MESH: Satellite and ML-based Estimation of Surface air quality at High resolution

S-MESH 2020 annual average

CAMS regional reanalysis 2020 annual average



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Method 3

ML-based downscaling of CAMS $PM_{2.5}$

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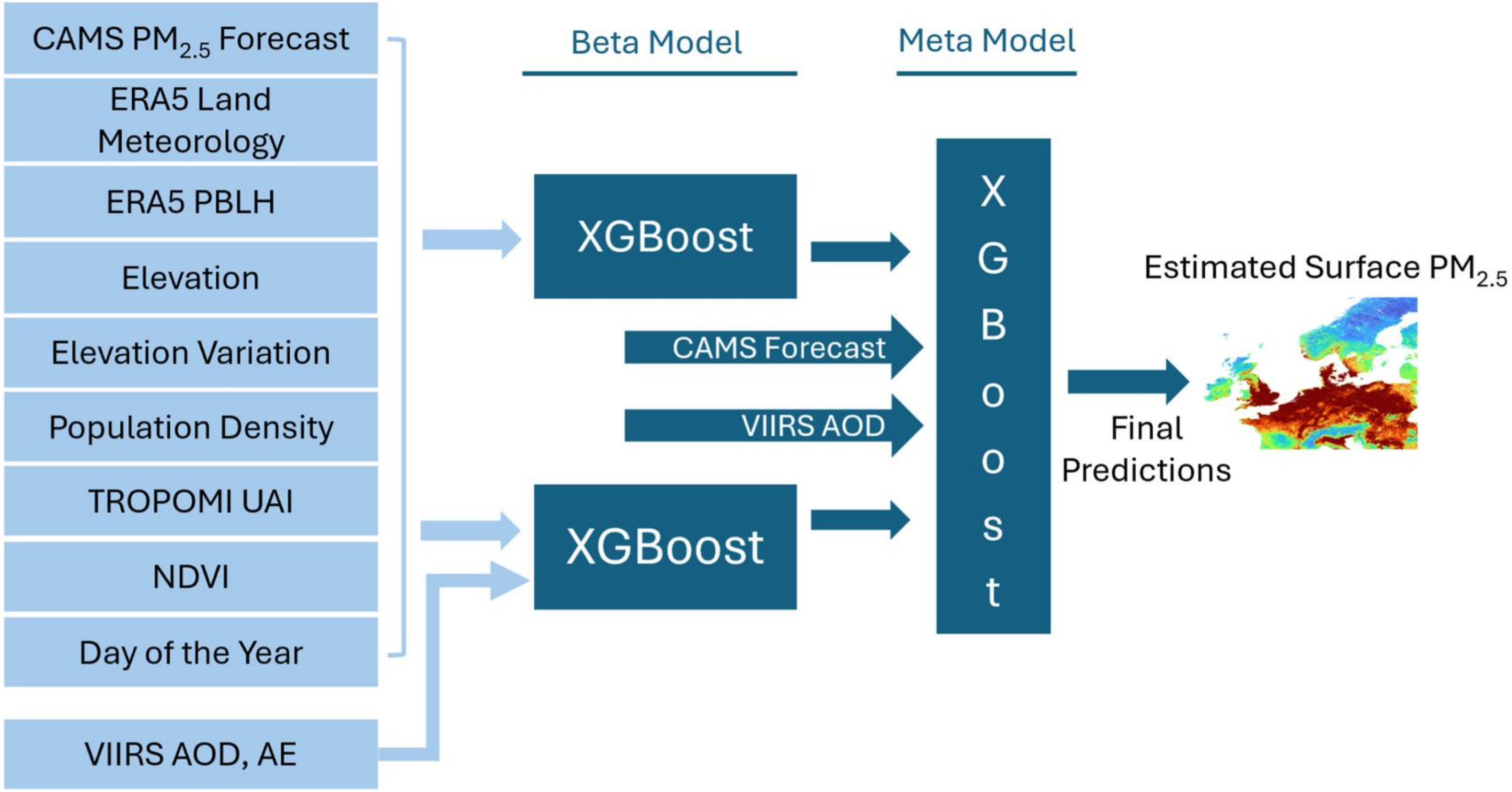
S-MESH for $PM_{2.5}$

Challenge
AOD is typically less directly linked to surface $PM_{2.5}$ than NO_2 TVCD is to surface NO_2.

Approach
Taking CAMS regional surface $PM_{2.5}$ as a foundation, train a stacked XGBoost model with VIIRS AOD and other variables against reference station observations.



Overall concept of **S-MESH for $PM_{2.5}$**. The innovative "Stacked ML" approach allows for dealing with spatial gaps in the VIIRS AOD and TROPOMI AI data and providing continuous gap-filled maps.

Shetty, S., Hamer, P. D., Stebel, K., Kylling, A., Hassani, A., Berntsen, T. K., & Schneider, P. (2025). Daily high-resolution surface $PM_{2.5}$ estimation over Europe by ML-based downscaling of the CAMS regional forecast. Environmental Research, 264, 120363.

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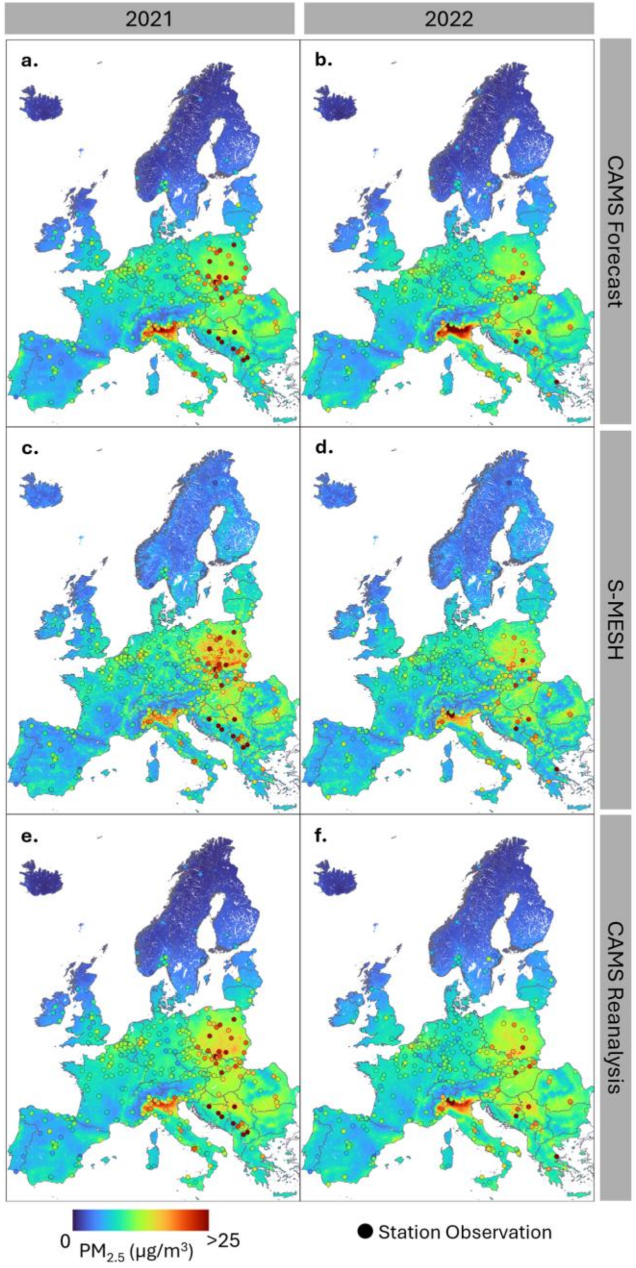

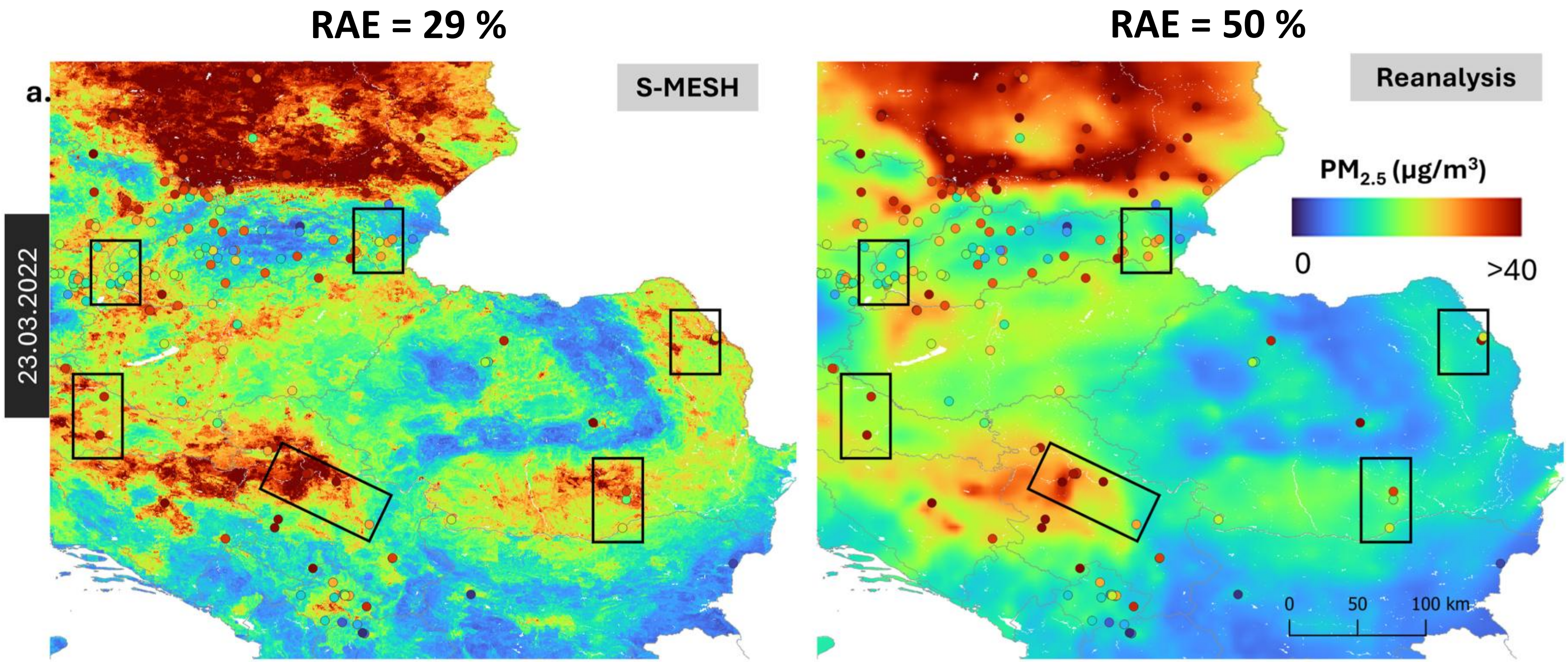

Daily average $PM_{2.5}$ concentrations (in µg/m^3) from S-MESH (left panel) and the CAMS reanalysis (right panel) for Poland, Hungary, Romania, and southeastern Europe for a day in a strong pollution episode (2022-03-23).

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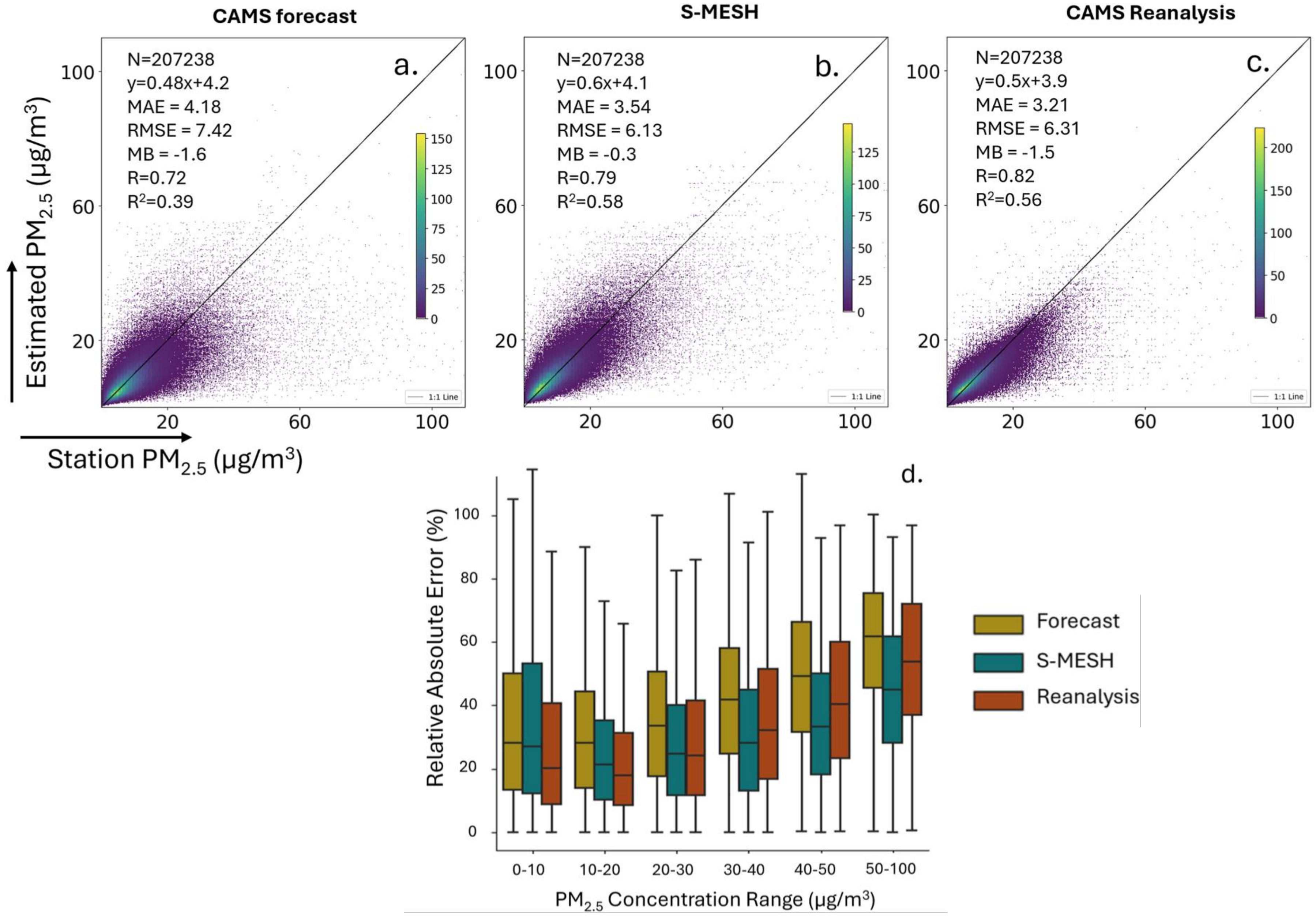

Overall, S-MESH for $PM_{2.5}$ produces **accuracy levels similar to the CAMS regional reanalysis** (and better for moderate/high pollution levels)

It can do so at **reduced computational expense**: Ca. 40 seconds for predicting a single day for all of Europe at 1 km.

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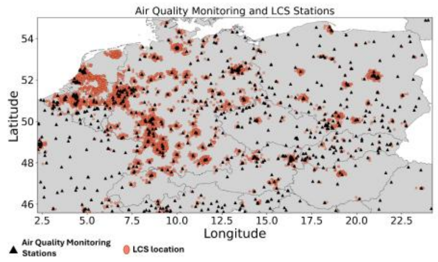
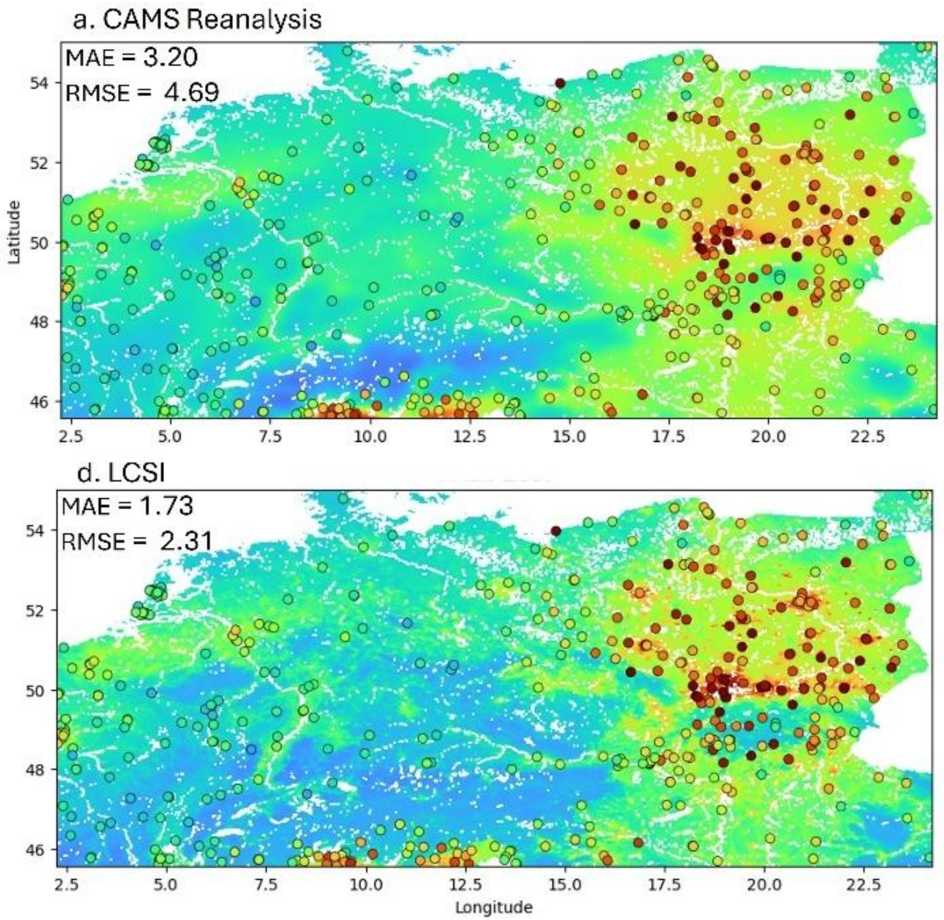
S-MESH: Integration of Earth Observation and Citizen Science



We integrate in S-MESH **quality-controlled** (Hassani et al., 2025) observations from **large-scale low-cost $PM_{2.5}$ sensor networks** (sensor.community and PurpleAir), providing high-density insitu measurements.

We find that a LCS-convolution layer as additional input to the stacked ML model shows **substantially improved accuracy over the CAMS reanalysis.**



CAMS Reanalysis

S-MESH with low-cost sensor convolution layer

Annual Average $PM_{2.5}$ levels for 2021 for a) CAMS regional reanalysis, d) S-MESH with low-cost sensors as convolution layer input. Point symbols indicate ref. stations.

Hassani, A., Salamalikis, V., Schneider, P., Stebel, K., & Castell, N. (2025). A scalable framework for harmonizing, standardization, and correcting crowd-sourced low-cost sensor PM2.5 data across Europe. Journal of Environmental Management, 380, 125100.

Shetty, S., Hassani, A., Salamalikis, V., Hamer, P. D., Stebel, K., Kylling, A., Castell, N., Berntsen, T. K., & Schneider, P. (2025). Large scale sensor networks improve continental scale PM2.5 monitoring. Manuscript in preparation for Environmental Research.

Summary

Converting satellite-based **vertically integrated AQ variables to surface concentrations** and **downscaling** them to finer resolution allows for **more direct EO use** for societal applications such as exposure assessment.

We use **three approaches** for this: 1) ”Classical” geostatistics for TROPOMI NO2, 2) ML-based for TROPOMI NO_2 and 3) ML-based for downscaling/correcting CAMS $PM_{2.5}$ using e.g. VIIRS AOD.

In particular the ML-based approaches show substantial promise, providing a **computationally very inexpensive alternative** that is roughly **equal to (NO_2) or outperforms ($PM_{2.5}$)** the **CAMS reanalysis**.

Outlook: Integrate hourly Sentinel-4 data in S-MESH

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More information

Contact Philipp Schneider (ps@nilu.no)

Websites citysatair.nilu.no
models.nilu.no/models/s-mesh/

References

Shetty, S., Schneider, P., Stebel, K., Hamer, P. D., Kylling, A., & Berntsen, T. K. (2024). Estimating surface NO2 concentrations over Europe using Sentinel-5P TROPOMI observations and Machine Learning. Remote Sensing of Environment, 312, 114321.

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European Space Agency

CitySatAir

Norwegian Space Agency

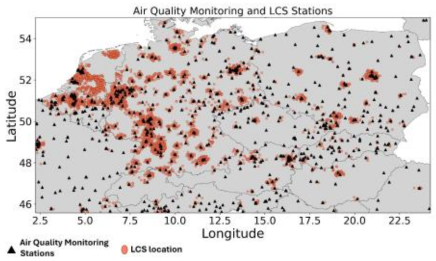
ECOMAP

European Commission

GreenEO

CitiObs

S-MESH: Integration of Earth Observation and Citizen Science



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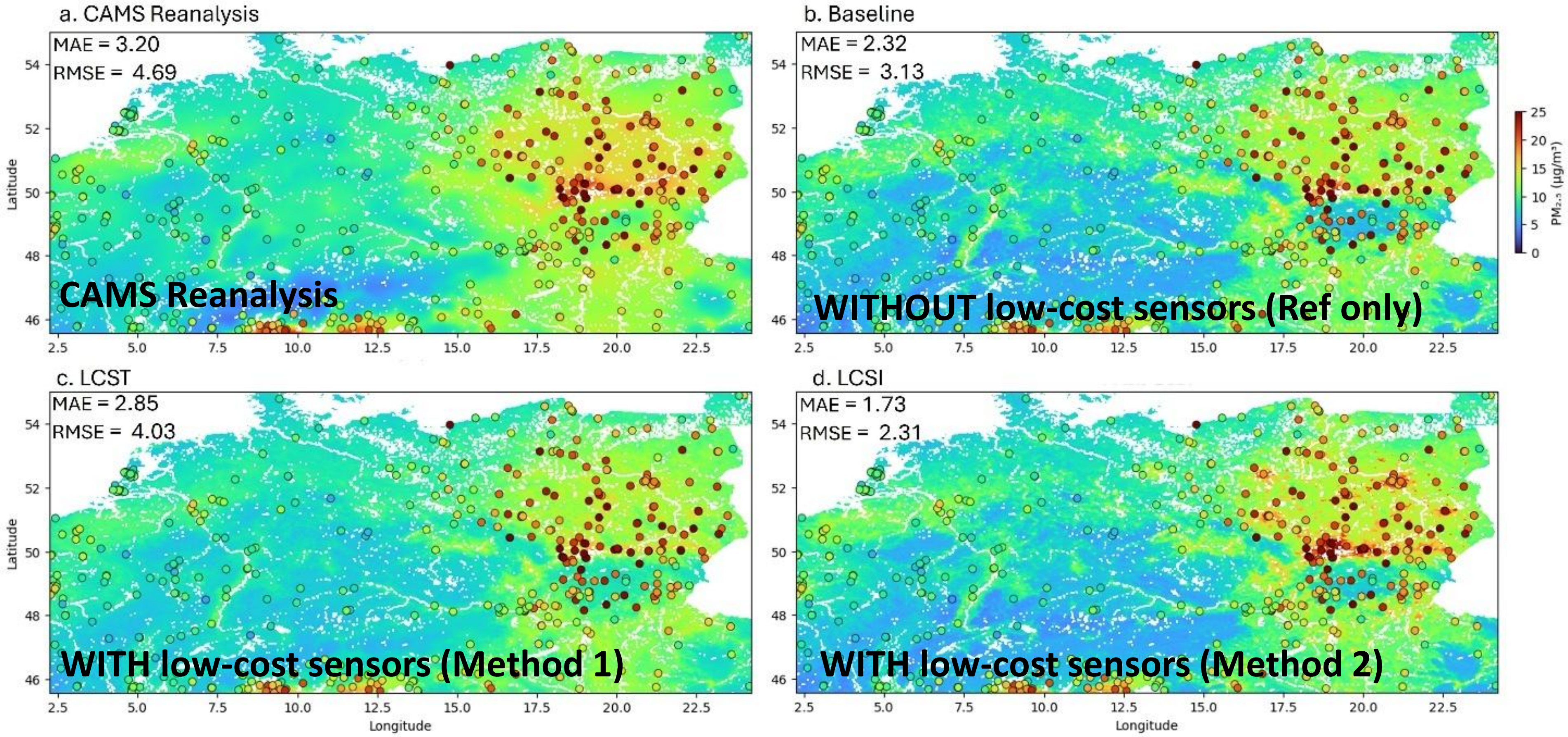

Annual Average $PM_{2.5}$ levels for 2021 for a) CAMS regional reanalysis, b) standard S-MESH, c) S-MESH with low-cost sensors as target, d) S-MESH with low-cost sensors as convolution layer input. Point symbols indicate ref. stations.

Hassani, A., Salamalikis, V., Schneider, P., Stebel, K., & Castell, N. (2025). A scalable framework for harmonizing, standardization, and correcting crowd-sourced low-cost sensor PM2.5 data across Europe. Journal of Environmental Management, 380, 125100.

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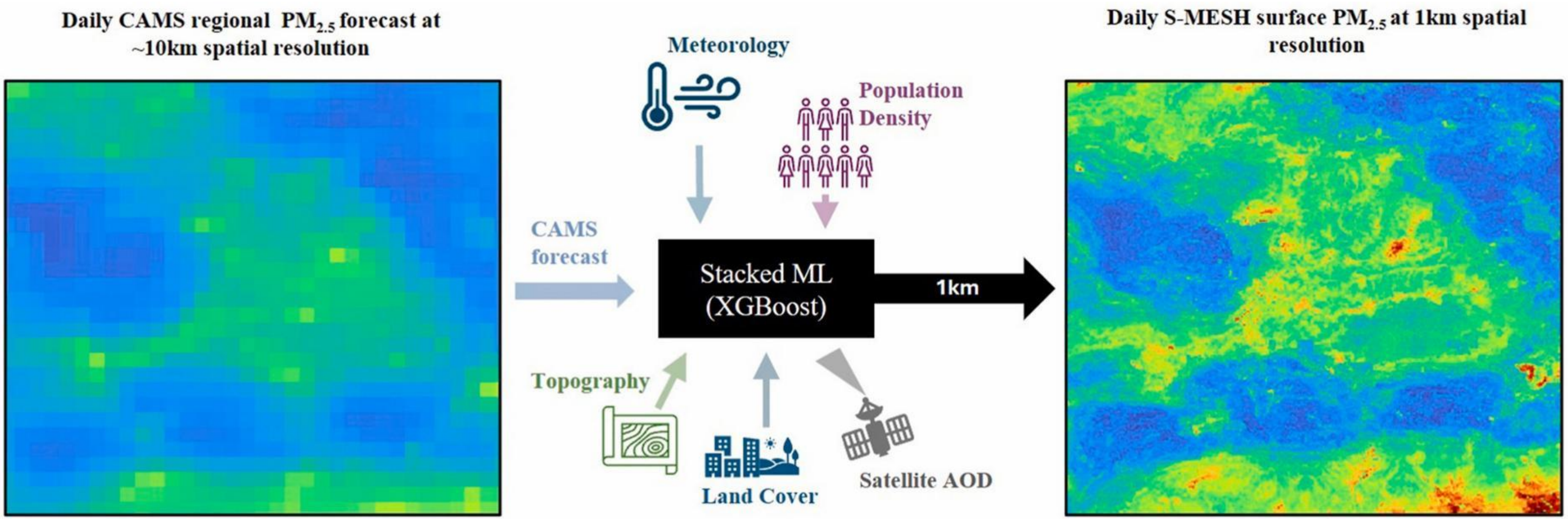
S-MESH: Synergy of Earth Observation and Machine Learning for Air Quality Monitoring in Europe

Simultaneous bias correction and downscaling of the CAMS regional ensemble



CAMS: RMSE = 7.42 µg/m3; MAE = 4.18 µg/m3; R^2 = 0.39

S-MESH: RMSE = 6.13 µg/m3; MAE = 3.54 µg/m3; R^2 = 0.58

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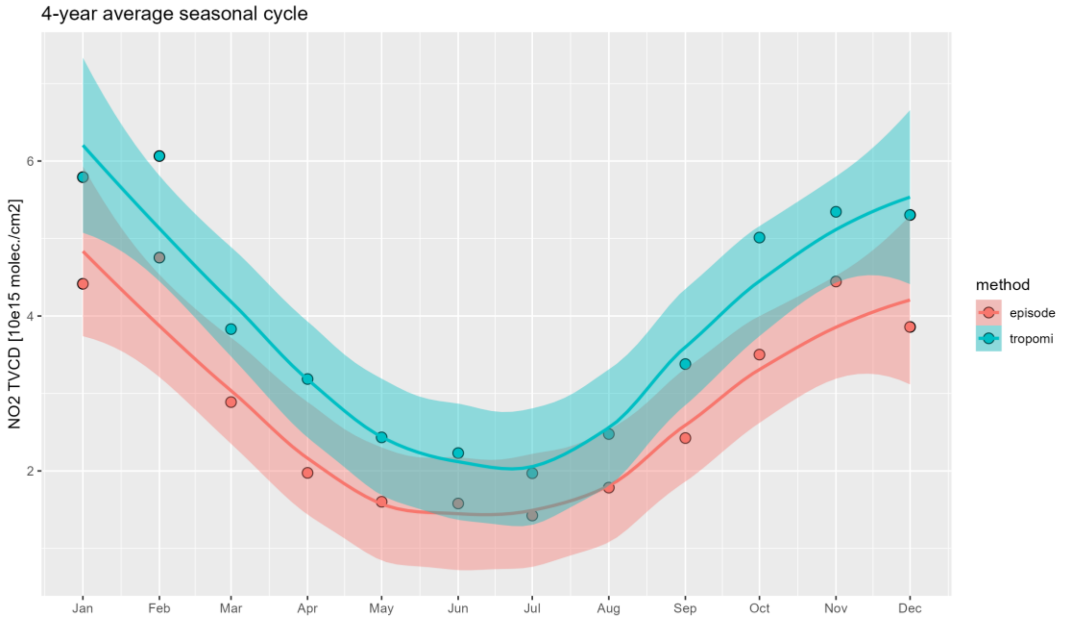
Linking EPISODE to TROPOMI NO2 TVCD

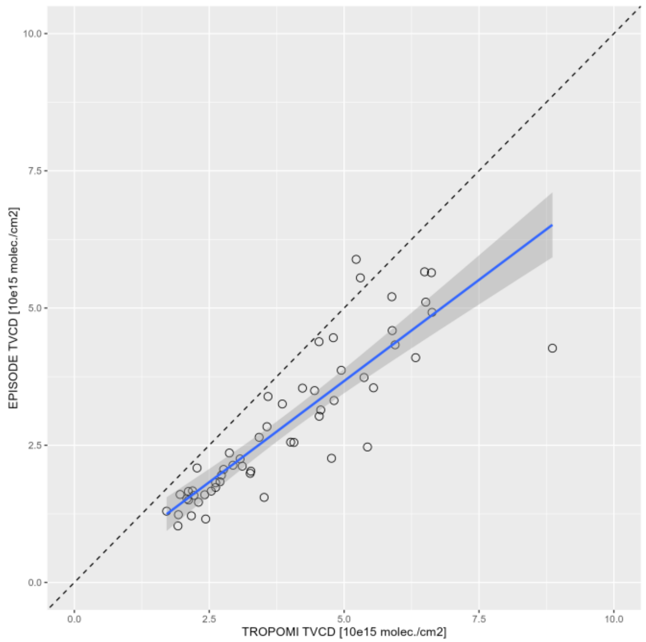
Processing steps

1. **Temporal interpolation** of the hourly EPISODE data to the exact TROPOMI overpass time.
2. Projection **to the CRS of the target geometry** and **spatially interpolated** to the irregular TROPOMI L2 pixel footprints with areal-weighting.
3. **Vertically interpolation** between EPISODE layers and TM5 layers
4. Application of the **TROPOMI L2 averaging kernel**
5. Calculation of **partial NO_2 columns** for each TM5 layer, and **vertical integration**

Outcome

The TVCD from EPISODE and TROPOMI show a **good correlation** overall but exhibit a **consistent bias**. The bias can be used to **adjust our bottom-up NOx emissions**.





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